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Long-Memory Amplifies Noisy Pseudogradients in DiLoCo

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In low-communication training regimes, communicating the necessary optimization state such as the parameters or gradients at every optimization step may be prohibitively expensive. A common alternative is to use *pseudogradient* methods, such as DiLoCo, which replace the per-step synchronized gradient update with a surrogate gradient computed from locally performed steps.

0.1. Setup and notation

To clarify notation, we use a tilde to denote random quantities. Any Latin or Greek symbol with a tilde (e.g., $\widetilde{x}$, $\widetilde{\theta}$) is a random variable defined on an underlying probability space $(\Omega, \mathcal{F}, \mathbb{P})$, for example $\widetilde{x} : \Omega \to \mathbb{R}^d$. A time-indexed sequence $(\widetilde{x}_k)_{k \geq 1}$ denotes a stochastic process. Symbols without a tilde denote constants or realizations.

0.2. Terminologies

Synchronized gradient. We use the term *faithful training* to refer to a baseline in which all workers synchronize at every optimization step so that the resulting parameter update coincides with the update that would be obtained under standard per-step data-parallel training on a single synchronized mini-batch. This includes settings that we are most familiar with, such as single-device training and data-parallel training with per-step synchronization (e.g., DDP/ FSDP-style training), where workers communicate the required optimization state at each step.

We denote the synchronized stochastic gradient produced under faithful training by

$$\nabla_\theta \ell(\widetilde{\theta}_k; \widetilde{\mathcal{B}_k})$$

This synchronized gradient is random because it depends on the randomly sampled mini-batch $\tilde{\mathcal{B}}$ via the data loader, and because the iterate $\widetilde{\theta}_k$ is determined by the training history, random initialization and the sequence of past mini-batch draws.

Pseudogradient. In low-communication regimes, per-step synchronization is prohibitively expensive due to resource constraints. DiLoCo mitigates this by running H local inner steps on each worker and then communicating a *parameter delta* — the difference between the shared parameters at the start of the round and the worker's parameters after H inner steps. These deltas are aggregated across workers and used as an outer-gradient surrogate by a global outer optimizer.

We denote the resulting DiLoCo pseudogradient by

$$\widetilde{\Delta}_k$$

This pseudogradient is random for the same underlying reasons, and additionally due to the local inner optimization mechanism and its implementation details used to generate the parameter delta.

In an ideal regime, we would have

$$\widetilde{\Delta}_k \approx \nabla_\theta \ell(\widetilde{\theta}_k; \widetilde{\mathcal{B}_k})$$

meaning that, despite operating under communication constraints, the pseudogradient update direction remains a close approximation to the per-step synchronized stochastic gradient, so that the resulting training trajectory remains comparable to the faithful baseline and will follow standard training practices that converge to a satisfactory solution.

A natural question is how perturbations $\widetilde{\epsilon}_k$ of the communicated pseudogradient propagate through the outer optimizer to produce the next iterate, and how much does this iterate deviate from the iterate obtained under per-step synchronized gradient updates.

Deviations may arise because each worker evaluates updates at parameters that have been locally modified, and because workers may observe different data shards and mini-batch sequences, likely optimizing a different loss landscape. Consequently, the surrogate update direction can differ from the synchronized gradient that would be obtained under per-step communication.

Formally, to assess the faithfulness of pseudogradients as proxies for synchronized gradient updates, we study how random *pseudogradient estimation error* (i.e., perturbations in the communicated update direction) affects the resulting parameter trajectory. Specifically, we formulate this as a *stochastic iterate-estimation problem*. We analyze how random perturbations caused by stochastic mini-batch sampling, algorithmic randomness, and possible systematic discrepancies influence the deviation between the pseudogradient-driven iterate and the faithful-training iterate.

We decompose the pseudogradient into an ideal component and an error component, e.g.,

$$\widetilde{\Delta}_k \;=\; \underbrace{\nabla_\theta \ell(\widetilde{\theta}_k; \widetilde{\mathcal{B}_k})}_{\text{ideal (synchronized) gradient}} \;+\; \underbrace{\tilde{\epsilon}_k}_{\text{pseudogradient estimation error}}$$

where $\widetilde{\epsilon}_k$ denotes the pseudogradient estimation error at optimization step k. This error may include both bias and stochastic variability, and it may be temporally correlated across steps, which we analyze in later sections.

In our analysis, we focus on the standard DiLoCo setup that uses Nesterov momentum in the outer optimizer and AdamW in the inner optimizer, and we use the theory to motivate our simple strategy of *periodic momentum restarts* whcih truncates the effective memory.

Let $\widetilde{\theta}_{t+1}^{\mathrm{DLC}} \in \mathbb{R}^d$ denote the DiLoCo outer iterate at time $t+1$, and let $\widetilde{\theta}_{t+1}^{\mathrm{S}} \in \mathbb{R}^d$ denote the corresponding per-step synchronized (faithful) outer iterate at the same time index, which serves as the target trajectory we aim to approximate. In practice, d is the model parameter dimension and is typically large.

Viewing $\widetilde{\theta}_{t+1}^{\mathrm{DLC}}$ as an estimator of $\widetilde{\theta}_{t+1}^{\mathrm{S}}$, we define the random iterate-estimation error $\widetilde{e}$ by

$$\underbrace{\widetilde{e}_{t+1}}_{\text{iterate-estimation error}} := \widetilde{\theta}_{t+1}^{\mathrm{DLC}} - \widetilde{\theta}_{t+1}^{\mathrm{S}}$$

For notational simplicity, we work at a fixed outer-round index $t \geq 1$, and when t is clear from context we write $\tilde{e} := \tilde{e}_{t+1} = \tilde{\theta}_{t+1}^{\mathrm{DLC}} - \tilde{\theta}_{t+1}^{S}$.

We proceed by deriving explicit representations for the mean and second-order statistics of $\widetilde{e}$ that expose the role of the long-memory coefficients $a_{k,t}$. We refer to these coefficients as *error amplifiers* and analyze their monotonicity properties in Section 2.1. We study both total-variance and directional projected-variance summaries to quantify how long-memory weighting in the outer momentum can amplify pseudogradient errors over time, even when such errors are initially small.

First, we derive an exact representation of the iterate-estimation error in terms of the per-step pseudogradient error. Specifically, defining $\widetilde{\epsilon}_k := \widetilde{\Delta}_k - \nabla_\theta \ell(\widetilde{\theta}_k; \widetilde{\mathcal{B}_k})$ and $\widetilde{e} := \widetilde{\theta}_{t+1}^{\mathrm{DLC}} - \widetilde{\theta}_{t+1}^{\mathrm{S}}$, we show that the iterate deviation can be written as a negatively scaled linear combination

$$\widetilde{e} = -\gamma \sum_{k=1}^{t} a_{k,t} \widetilde{\epsilon}_k$$

This identity in Lemma 2 isolates the source of iterate deviation as the accumulation of pseudogradient errors weighted by the long-memory coefficients $a_{k,t}$.

Second, taking expectations of the above identity yields Lemma 3, which makes explicit how any nonzero mean error (bias) can accumulate through the same long-memory weighting.

Third, to quantify stochastic deviations, we derive an explicit covariance expansion for the iterate-estimation error in Lemma 4

$$\mathrm{Cov}[\widetilde{e}] = \gamma^2 \sum_{k=1}^{t} \sum_{l=1}^{t} a_{k,t} a_{l,t} \, \mathrm{Cov}(\widetilde{\epsilon}_l, \widetilde{\epsilon}_k)$$

This form exposes the pair of temporal error-amplification factors $a_{k,t} a_{l,t}$, i.e., products of the *error amplifiers* $a_{k,t}$, and shows how temporal cross-covariances of the error process enter the iterate covariance.

1. Second-order statistics of the iterate-estimation error

From the covariance expansion in Lemma 4, we study multiple variance-type summaries under worst-case control. We first consider the standard total variance of a random vector, given by the trace

$$\mathrm{tr}(\mathrm{Cov}[\widetilde{e}])$$

and express it as a sum of self-covariance and cross-covariance contributions in Corollary 5. Since cross-covariance terms may take either sign, we separately provide a general upper bound in Corollary 6 that controls the total variance in terms of operator norms of the cross-covariance matrices and the error amplification coefficients $a_{k,t} a_{l,t}$.

For completeness, we also analyze the special case in which the temporal cross-covariances vanish in Corollary 7, i.e., $\mathrm{Cov}(\widetilde{\epsilon}_l, \widetilde{\epsilon}_k) = 0_{d\times d}$ for $k \neq l$. Under this assumption, the total variance collapses to its diagonal-only contribution, which we denote by V.

$$\mathrm{tr}(\mathrm{Cov}[\widetilde{e}]) = \mathrm{V} := \gamma^2 \sum_{k=1}^{t} a_{k,t}^2 \, \mathrm{tr}\left(\mathrm{Cov}(\widetilde{\epsilon}_k, \widetilde{\epsilon}_k)\right)$$

This isolates the amplification effect through the squared error-amplifier coefficients $a_{k,t}^2$, and we also provide a corresponding upper bound on V in Corollary 8.

We next consider a fully general directional measure of deviation via projected variance. For any unit direction $u \in \mathbb{R}^d$ with $\|u\|_2 = 1$, we study $\mathrm{Var}[u^\top \widetilde{e}_{t+1}]$ and show that it can be packaged as a quadratic form in a stacked vector of coefficients, yielding the bound in Theorem 9.

$$\mathrm{Var}[u^\top \widetilde{e}] \leq \gamma^2 \|\hat{\Sigma}\|_{\mathrm{op}} \|a_t\|_2^2$$

where $\hat{\Sigma}$ is the stacked covariance matrix of the error sequence $(\widetilde{\epsilon}_k)_{k=1}^t$ and $a_t := [a_{1,t}, \ldots, a_{t,t}]^\top \in \mathbb{R}^t$ collects the error-amplifier coefficients. This bound again highlights $\|a_t\|_2^2$ as a global amplification factor induced by long-memory weighting.

Finally, we justify the terminology *error amplifiers* by analyzing the coefficients $a_{k,t}$ themselves. Using the closed-form expression for $a_{k,t}$, we show in Theorem 10 that $a_{k,t}$ varies monotonically with the time gap $t-k$. This makes precise how older memory terms can receive larger effective weights in the unrolled outer update, thereby amplifying the contribution of earlier pseudogradient errors over long horizons.

While our derivation is specific to the Nesterov outer update used in DiLoCo, the same mechanism is expected to arise more broadly in long-memory momentum schemes whose unrolled coefficients exhibit different forms of geometric weighting.

2. Proofs

Lemma 1 *Closed-form update rule for Nesterov momentum (1-based indexing) via induction. Let $\gamma \in (0, \infty)$ denote the learning rate, let $\tau \in [0,1]$ denote the dampening parameter, and let $\mu \in [0,1)$ denote the momentum parameter. Let θ_1 be the initial iterate. For $t \geq 1$, define the momentum recursion*

$$m_{t+1} = \mu m_t + (1-\tau) g_t \tag{1}$$

and the iterate update

$$\theta_{t+1} = \theta_t - \gamma\big(g_t + \mu m_{t+1}\big) \tag{2}$$

as in the PyTorch Nesterov momentum convention **?**. *Then for all $t \geq 1$,*

$$\theta_{t+1} = \theta_1 - \gamma \sum_{k=1}^{t} \left(1 + \frac{(1-\tau)\big(\mu - \mu^{t-k+2}\big)}{1-\mu}\right) g_k - \gamma \mu^2 \frac{1-\mu^t}{1-\mu} m_1 \tag{3}$$

Remark. In our training setting, g_k will later be instantiated as a stochastic gradient, but no probabilistic assumptions are made for the closed-form unrolling.

Proof We first unroll the momentum recurrence to obtain a closed-form expression for m_{t+1} by induction.

Claim. For all integers $t \geq 1$,

$$m_{t+1} = \mu^t m_1 + \sum_{k=1}^{t} \mu^{t-k} (1-\tau) g_k \tag{4}$$

Base case ($t = 1$). By the recursion $m_{t+1} = \mu m_t + (1-\tau) g_t$ we have

$$m_2 = \mu m_1 + (1-\tau) g_1 \tag{5}$$

and the claimed right-hand side equals $\mu^1 m_1 + \sum_{k=1}^{1} \mu^{1-k}(1-\tau) g_k = \mu m_1 + (1-\tau) g_1$

Induction step. Fix $t \geq 1$ and assume as the induction hypothesis that m_t holds,

$$m_t = \mu^{t-1} m_1 + \sum_{k=1}^{t-1} \mu^{(t-1)-k} (1-\tau) g_k \tag{6}$$

Using the momentum recursion,

$$m_{t+1} = \mu m_t + (1-\tau) g_t \tag{7}$$

and substituting the induction hypothesis for m_t gives

$$m_{t+1} = \mu\Big(\mu^{t-1}m_1 + \sum_{k=1}^{t-1} \mu^{(t-1)-k}(1-\tau)g_k\Big) + (1-\tau)g_t \tag{8}$$

$$= \mu^t m_1 + \sum_{k=1}^{t-1} \mu^{t-k}(1-\tau)g_k + \mu^0(1-\tau)g_t \tag{9}$$

$$= \mu^t m_1 + \sum_{k=1}^{t} \mu^{t-k}(1-\tau)g_k \tag{10}$$

which is exactly our claim for m_{t+1}.

We now unroll the parameter recurrence to obtain a closed-form expression for θ_{t+1} by induction, using 1-based forward indexing.

Claim. For all integers $t \geq 1$,

$$\theta_{t+1} = \theta_1 - \sum_{k=1}^{t} \gamma\big(g_k + \mu m_{k+1}\big) \tag{11}$$

Base case ($t = 1$). By the recursion $\theta_{t+1} = \theta_t - \gamma\big(g_t + \mu m_{t+1}\big)$ we have

$$\theta_2 = \theta_1 - \gamma\big(g_1 + \mu m_2\big) \tag{12}$$

and the claimed right-hand side equals $\theta_1 - \sum_{k=1}^{1} \gamma\big(g_k + \mu m_{k+1}\big) = \theta_1 - \gamma\big(g_1 + \mu m_2\big)$.

Induction step. Fix $t \geq 1$ and assume as the induction hypothesis that

$$\theta_t = \theta_1 - \sum_{k=1}^{t-1} \gamma\big(g_k + \mu m_{k+1}\big) \tag{13}$$

Using the update recursion,

$$\theta_{t+1} = \theta_t - \gamma\big(g_t + \mu m_{t+1}\big) \tag{14}$$

and substituting the induction hypothesis for θ_t gives

$$\theta_{t+1} = \Big(\theta_1 - \sum_{k=1}^{t-1} \gamma\big(g_k + \mu m_{k+1}\big)\Big) - \gamma\big(g_t + \mu m_{t+1}\big) \tag{15}$$

$$= \theta_1 - \sum_{k=1}^{t} \gamma\big(g_k + \mu m_{k+1}\big) \tag{16}$$

which is exactly our desired identity for θ_{t+1} in the claim.

Substituting the unrolling of m_{k+1} into the unrolling of θ_{t+1} gives

$$\theta_{t+1} = \theta_1 - \sum_{k=1}^{t} \gamma\Big(g_k + \mu\Big(\mu^k m_1 + \sum_{s=1}^{k} \mu^{k-s}(1-\tau)g_s\Big)\Big) \tag{17}$$

$$= \theta_1 - \sum_{k=1}^{t} \gamma\Big(g_k + \mu^{k+1} m_1 + (1-\tau)\sum_{s=1}^{k} \mu^{k-s+1} g_s\Big) \tag{18}$$

$$= \theta_1 - \sum_{k=1}^{t} \gamma g_k - \sum_{k=1}^{t} \gamma \mu^{k+1} m_1 - \gamma(1-\tau)\sum_{k=1}^{t}\sum_{s=1}^{k} \mu^{k-s+1} g_s \tag{19}$$

$$= \theta_1 - \gamma\sum_{k=1}^{t} g_k - \gamma\sum_{k=1}^{t} \mu^{k+1} m_1 - \gamma(1-\tau)\sum_{k=1}^{t}\sum_{s=1}^{k} \mu^{k-s+1} g_s \tag{20}$$

By reindexing the triangular double sum to group terms by g_l (graphically, this corresponds to summing over the same triangular index set by columns instead of by rows), we expose an inner geometric series in μ, and we obtain

$$\theta_{t+1} = \theta_1 - \gamma\sum_{k=1}^{t} g_k - \Big(\gamma\sum_{k=1}^{t} \mu^{k+1}\Big) m_1 - \gamma(1-\tau)\sum_{l=1}^{t}\sum_{k=l}^{t} \mu^{k-l+1} g_l \tag{21}$$

$$= \theta_1 - \gamma\sum_{k=1}^{t} g_k - \Big(\gamma\sum_{k=1}^{t} \mu^{k+1}\Big) m_1 - \gamma(1-\tau)\sum_{l=1}^{t}\Big(\sum_{r=1}^{t-l+1} \mu^{r}\Big) g_l \tag{22}$$

Since these terms are geometric sums when $|\mu| < 1$, we can express them as

$$\gamma\sum_{k=1}^{t} \mu^{k+1} = \gamma\mu\sum_{k=1}^{t} \mu^{k} \tag{23}$$

$$= \gamma\mu\Big(\Big(\sum_{k=0}^{t} \mu^{k}\Big) - \mu^0\Big) \tag{24}$$

$$= \gamma\mu\Big(\frac{1-\mu^{t+1}}{1-\mu} - 1\Big) \tag{25}$$

$$= \gamma\mu\Big(\frac{1-\mu^{t+1} - (1-\mu)}{1-\mu}\Big) \tag{26}$$

$$= \gamma\mu\Big(\frac{\mu-\mu^{t+1}}{1-\mu}\Big) \tag{27}$$

$$= \gamma\mu^2\Big(\frac{1-\mu^{t}}{1-\mu}\Big) \tag{28}$$

$$\sum_{r=1}^{t-l+1} \mu^r = \Big(\sum_{r=0}^{t-l+1} \mu^r \Big) - \mu^0 \tag{29}$$

$$= \Big(\frac{1-\mu^{t-l+2}}{1-\mu} \Big) - 1 \tag{30}$$

$$= \frac{1-\mu^{t-l+2}}{1-\mu} - \frac{1-\mu}{1-\mu} \tag{31}$$

$$= \frac{\mu-\mu^{t-l+2}}{1-\mu} \tag{32}$$

Substituting the geometric-sum identities, we obtain

$$\theta_{t+1} = \theta_1 - \gamma \sum_{k=1}^{t} g_k - \gamma\mu^2 \Big(\frac{1-\mu^t}{1-\mu} \Big) m_1 - \gamma(1-\tau) \sum_{l=1}^{t} \frac{\mu-\mu^{t-l+2}}{1-\mu} g_l \tag{33}$$

$$= \theta_1 - \gamma \sum_{k=1}^{t} g_k - \gamma(1-\tau) \sum_{l=1}^{t} \frac{\mu-\mu^{t-l+2}}{1-\mu} g_l - \gamma\mu^2 \Big(\frac{1-\mu^t}{1-\mu} \Big) m_1 \tag{34}$$

$$= \theta_1 - \gamma \left(\sum_{k=1}^{t} g_k + (1-\tau) \sum_{l=1}^{t} \frac{\mu-\mu^{t-l+2}}{1-\mu} g_l \right) - \gamma\mu^2 \Big(\frac{1-\mu^t}{1-\mu} \Big) m_1 \tag{35}$$

Since both summations are indexed over the same gradient sequence, we can relabel l as k in the second sum and regroup terms.

$$\theta_{t+1} = \theta_1 - \gamma \sum_{k=1}^{t} \left(1 + \frac{(1-\tau)(\mu-\mu^{t-k+2})}{1-\mu} \right) g_k - \gamma\mu^2 \Big(\frac{1-\mu^t}{1-\mu} \Big) m_1 \tag{36}$$

■

Lemma 2 (Iterate estimation error) *Let $\widetilde{\theta}_{t+1}^{DLC} \in \mathbb{R}^d$ denote the DiLoCo outer iterate at time $t+1$, and let $\widetilde{\theta}_{t+1}^{S} \in \mathbb{R}^d$ denote the corresponding per-step synchronized (faithful) outer iterate at the same time index. Here θ_1 and m_1 denote the common initial state of the outer-update window under consideration, and use the same outer Nesterov hyperparameters γ, μ, τ, where the unrolled coefficients $a_{k,t}$ and c_t apply to both. Define the scalar coefficients*

$$a_{k,t} := \left(1 + \frac{(1-\tau)\big(\mu-\mu^{t-k+2}\big)}{1-\mu} \right), \qquad c_t := \mu^2 \left(\frac{1-\mu^t}{1-\mu} \right) \tag{37}$$

and define the per-step pseudogradient error vector

$$\widetilde{\epsilon}_k := \widetilde{\Delta}_k^{DLC} - \nabla_\theta \ell(\widetilde{\theta}_k; \widetilde{\mathcal{B}_k}) \in \mathbb{R}^d \tag{38}$$

Viewing $\widetilde{\theta}_{t+1}^{DLC}$ as an estimator of the per-step synchronized iterate $\widetilde{\theta}_{t+1}^{S}$, we define the iterate-estimation error *as $\widetilde{e} := \widetilde{\theta}_{t+1}^{DLC} - \widetilde{\theta}_{t+1}^{S}$. Then $\widetilde{e}$ can be written as*

$$\widetilde{e} = -\gamma \sum_{k=1}^{t} a_{k,t} \widetilde{\epsilon}_k \tag{39}$$

Proof We write the closed-form Nesterov outer-update for DiLoCo and for faithful training in the following manner.

For compactness, let $a_{k,t}$ denote the scalar coefficient multiplying the k-th update direction in the unrolled Nesterov recursion, and let c_t denote the coefficient multiplying the initial momentum term. Specifically,

$$a_{k,t} := \left(1 + \frac{(1-\tau)\big(\mu - \mu^{t-k+2}\big)}{1-\mu}\right) \tag{40}$$

$$c_t := \mu^2 \left(\frac{1-\mu^t}{1-\mu}\right) \tag{41}$$

We denote the DiLoCo pseudogradient at step k by $\widetilde{\Delta}_k^{DLC} \in \mathbb{R}^d$ and the per-step synchronized (faithful) stochastic gradient at step k by $\nabla_\theta \ell(\widetilde{\theta}_k; \widetilde{\mathcal{B}_k}) \in \mathbb{R}^d$. In the stochastic training setting, both quantities may be random due to mini-batch sampling and the induced randomness of the iterate.

The corresponding unrolled outer updates are then

$$\widetilde{\theta}_{t+1}^{\mathrm{DLC}} = \theta_1 - \gamma \sum_{k=1}^{t} a_{k,t} \widetilde{\Delta}_k^{DLC} - \gamma c_t m_1 \tag{42}$$

$$\text{(DiLoCo outer iterate at time } t+1\text{)} \tag{43}$$

$$\widetilde{\theta}_{t+1}^{\mathrm{S}} = \theta_1 - \gamma \sum_{k=1}^{t} a_{k,t} \nabla_\theta \ell(\widetilde{\theta}_k; \widetilde{\mathcal{B}_k}) - \gamma c_t m_1 \tag{44}$$

$$\text{(Per-step synchronized outer iterate at time } t+1\text{)} \tag{45}$$

We now take the difference between the two trajectories to isolate the source of their deviation.

$$\begin{aligned}
\widetilde{\theta}_{t+1}^{\mathrm{DLC}} - \widetilde{\theta}_{t+1}^{\mathrm{S}} &= \Big(\theta_1 - \gamma \sum_{k=1}^{t} a_{k,t} \widetilde{\Delta}_k^{DLC} - \gamma c_t m_1\Big) - \Big(\theta_1 - \gamma \sum_{k=1}^{t} a_{k,t} \nabla_\theta \ell(\widetilde{\theta}_k; \widetilde{\mathcal{B}_k}) - \gamma c_t m_1\Big) \\
&= -\gamma \sum_{k=1}^{t} a_{k,t} \widetilde{\Delta}_k^{DLC} + \gamma \sum_{k=1}^{t} a_{k,t} \nabla_\theta \ell(\widetilde{\theta}_k; \widetilde{\mathcal{B}_k}) \\
&= -\gamma \left(\sum_{k=1}^{t} a_{k,t} \widetilde{\Delta}_k^{DLC} - \sum_{k=1}^{t} a_{k,t} \nabla_\theta \ell(\widetilde{\theta}_k; \widetilde{\mathcal{B}_k}) \right) \quad \text{(factor out } -\gamma\text{)} \\
&= -\gamma \sum_{k=1}^{t} a_{k,t} \big(\widetilde{\Delta}_k^{DLC} - \nabla_\theta \ell(\widetilde{\theta}_k; \widetilde{\mathcal{B}_k}) \big) \quad \text{(by regrouping)}
\end{aligned}$$

We let $\widetilde{\epsilon}_k \in \mathbb{R}^d$ denote the *pseudogradient error* at step k.

$$\widetilde{\epsilon}_k := \widetilde{\Delta}_k^{DLC} - \nabla_\theta \ell(\widetilde{\theta}_k; \widetilde{\mathcal{B}_k}) \tag{46}$$

Then we can express the iterate difference compactly as

$$\widetilde{\theta}_{t+1}^{\mathrm{DLC}} - \widetilde{\theta}_{t+1}^{\mathrm{S}} = -\gamma \sum_{k=1}^{t} a_{k,t} \widetilde{\epsilon}_k \quad \text{(by substitution and regrouping)} \tag{47}$$

Define the iterate difference random vector (the *iterate-estimation error*) by $\widetilde{e} := \widetilde{\theta}_{t+1}^{\mathrm{DLC}} - \widetilde{\theta}_{t+1}^{\mathrm{S}}$, so that, equivalently,

$$\widetilde{e} = -\gamma \sum_{k=1}^{t} a_{k,t} \widetilde{\epsilon}_k \tag{48}$$

■

Remarks. The above identity implies that, for each realization of the randomness, $\widetilde{e}$ lies in the linear span of the per-step pseudogradient errors, i.e.,

$$\widetilde{e} \in \mathrm{span}\big(\{\widetilde{\epsilon}_1, \ldots, \widetilde{\epsilon}_t\}\big).$$

Consequently, we can study the geometry of the error sequence $\{\widetilde{\epsilon}_k\}_{k=1}^{t}$ to understand which directions the DiLoCo trajectory can potentially deviate from the faithful trajectory.

Lemma 3 (Expectation of iterate-estimation error) *Let $\widetilde{e} := \widetilde{\theta}_{t+1}^{DLC} - \widetilde{\theta}_{t+1}^{S} \in \mathbb{R}^d$ denote the iterate-estimation error, where $\widetilde{\theta}_{t+1}^{DLC}$ and $\widetilde{\theta}_{t+1}^{S}$ are defined as in Lemma 2. Then*

$$\mathbb{E}[\widetilde{e}] = -\gamma \sum_{k=1}^{t} a_{k,t} \mathbb{E}[\widetilde{\epsilon}_k] \tag{49}$$

Proof We compute the mean deviation between the DiLoCo iterate and the per-step synchronized iterate.

For compactness, define the iterate-estimation error random vector

$$\widetilde{e} := \widetilde{\theta}_{t+1}^{\mathrm{DLC}} - \widetilde{\theta}_{t+1}^{\mathrm{S}} \in \mathbb{R}^d$$

Then

$$\mathbb{E}[\widetilde{e}] = \mathbb{E}\Big[\widetilde{\theta}_{t+1}^{\mathrm{DLC}} - \widetilde{\theta}_{t+1}^{\mathrm{S}}\Big] \tag{50}$$

$$= \mathbb{E}\Big[-\gamma \sum_{k=1}^{t} a_{k,t} \widetilde{\epsilon}_k\Big] \quad (\text{by substituting Lemma 2}) \tag{51}$$

$$= -\gamma \sum_{k=1}^{t} a_{k,t} \mathbb{E}[\widetilde{\epsilon}_k] \quad (\text{by linearity of expectation}) \tag{52}$$

■

Remarks. In general, $\mathbb{E}[\widetilde{\epsilon}_k]$ need not be zero. This is because the DiLoCo pseudogradient is formed from local training trajectories with locally updated parameters and potentially different data shards induced by random data partitioning and mini-batch sampling, which can induce discrepancy relative to the per-step synchronized gradient and hence yield $\mathbb{E}[\widetilde{\epsilon}_k] \neq 0$.

Lemma 4 (Covariance of iterate-estimation error) *Let $\widetilde{e} := \widetilde{\theta}_{t+1}^{DLC} - \widetilde{\theta}_{t+1}^{S} \in \mathbb{R}^d$ denote the iterate-estimation error, by Lemma 2, we can express $\widetilde{e}$ as*

$$\widetilde{e} = -\gamma \sum_{k=1}^{t} a_{k,t} \widetilde{\epsilon}_k$$

where $\widetilde{\epsilon}_k \in \mathbb{R}^d$ denotes the pseudogradient error at step k. Then

$$\operatorname{Cov}[\widetilde{e}] = \gamma^2 \sum_{k=1}^{t} \sum_{l=1}^{t} a_{k,t} a_{l,t} \operatorname{Cov}(\widetilde{\epsilon}_l, \widetilde{\epsilon}_k) \tag{53}$$

where $\operatorname{Cov}(\widetilde{\epsilon}_l, \widetilde{\epsilon}_k) := \mathbb{E}\big[(\widetilde{\epsilon}_k - \mathbb{E}[\widetilde{\epsilon}_k])(\widetilde{\epsilon}_l - \mathbb{E}[\widetilde{\epsilon}_l])^\top\big] \in \mathbb{R}^{d\times d}$ denotes the cross-covariance matrix between $\widetilde{\epsilon}_k$ and $\widetilde{\epsilon}_l$, where the subscripts k and l index the corresponding optimization time steps.

Proof We can then express $\operatorname{Cov}[\widetilde{e}]$ by definition as

$$\begin{aligned}
\operatorname{Cov}[\widetilde{e}] &= \mathbb{E}\big[(\widetilde{e} - \mathbb{E}[\widetilde{e}])(\widetilde{e} - \mathbb{E}[\widetilde{e}])^\top\big] \\
&= \mathbb{E}\Big[\Big(-\gamma \sum_{l=1}^{t} a_{l,t}\widetilde{\epsilon}_l + \gamma \sum_{l=1}^{t} a_{l,t}\mathbb{E}[\widetilde{\epsilon}_l]\Big)\Big(-\gamma \sum_{k=1}^{t} a_{k,t}\widetilde{\epsilon}_k + \gamma \sum_{k=1}^{t} a_{k,t}\mathbb{E}[\widetilde{\epsilon}_k]\Big)^\top\Big] \\
&\quad (\text{ by substituting Lemma 2 and Lemma 3 }) \\
&= \mathbb{E}\Big[\Big(-\gamma \sum_{l=1}^{t} a_{l,t}\big(\widetilde{\epsilon}_l - \mathbb{E}[\widetilde{\epsilon}_l]\big)\Big)\Big(-\gamma \sum_{k=1}^{t} a_{k,t}\big(\widetilde{\epsilon}_k - \mathbb{E}[\widetilde{\epsilon}_k]\big)\Big)^\top\Big] \quad (\text{ by regrouping }) \\
&= \mathbb{E}\Big[(-\gamma)(-\gamma)\Big(\sum_{l=1}^{t} a_{l,t}\big(\widetilde{\epsilon}_l - \mathbb{E}[\widetilde{\epsilon}_l]\big)\Big)\Big(\sum_{k=1}^{t} a_{k,t}\big(\widetilde{\epsilon}_k - \mathbb{E}[\widetilde{\epsilon}_k]\big)\Big)^\top\Big] \quad (\text{ by homogeneity }) \\
&= \mathbb{E}\Big[\gamma^2\Big(\sum_{l=1}^{t} a_{l,t}\big(\widetilde{\epsilon}_l - \mathbb{E}[\widetilde{\epsilon}_l]\big)\Big)\Big(\sum_{k=1}^{t} a_{k,t}\big(\widetilde{\epsilon}_k - \mathbb{E}[\widetilde{\epsilon}_k]\big)\Big)^\top\Big] \\
&= \mathbb{E}\Big[\gamma^2\Big(\sum_{l=1}^{t} a_{l,t}\big(\widetilde{\epsilon}_l - \mathbb{E}[\widetilde{\epsilon}_l]\big)\Big)\Big(\sum_{k=1}^{t} a_{k,t}\big(\widetilde{\epsilon}_k - \mathbb{E}[\widetilde{\epsilon}_k]\big)^\top\Big)\Big] \quad (\text{ by linearity of transpose }) \\
&= \mathbb{E}\Big[\gamma^2 \sum_{k=1}^{t} a_{k,t}\Big(\sum_{l=1}^{t} a_{l,t}\big(\widetilde{\epsilon}_l - \mathbb{E}[\widetilde{\epsilon}_l]\big)\Big)\big(\widetilde{\epsilon}_k - \mathbb{E}[\widetilde{\epsilon}_k]\big)^\top\Big] \quad (\text{ distribute the sum over } k\text{ }) \\
&= \mathbb{E}\Big[\gamma^2 \sum_{k=1}^{t}\sum_{l=1}^{t} a_{k,t}a_{l,t}\big(\widetilde{\epsilon}_l - \mathbb{E}[\widetilde{\epsilon}_l]\big)\big(\widetilde{\epsilon}_k - \mathbb{E}[\widetilde{\epsilon}_k]\big)^\top\Big] \quad (\text{ by distributivity }) \\
&= \gamma^2 \sum_{k=1}^{t}\sum_{l=1}^{t} a_{k,t}a_{l,t}\mathbb{E}\Big[\big(\widetilde{\epsilon}_l - \mathbb{E}[\widetilde{\epsilon}_l]\big)\big(\widetilde{\epsilon}_k - \mathbb{E}[\widetilde{\epsilon}_k]\big)^\top\Big] \quad (\text{ by linearity of expectation }) \\
&= \gamma^2 \sum_{k=1}^{t}\sum_{l=1}^{t} a_{k,t}a_{l,t}\operatorname{Cov}(\widetilde{\epsilon}_l, \widetilde{\epsilon}_k) \quad (\text{ by the definition of cross-covariance matrix })
\end{aligned}$$

Note. Because $\widetilde{\epsilon}_k$ and $\widetilde{\epsilon}_l$ are random vectors in the same coordinate space $\mathbb{R}^d$, the cross-covariance matrix $\operatorname{Cov}(\widetilde{\epsilon}_l, \widetilde{\epsilon}_k)$ is a square matrix in $\mathbb{R}^{d\times d}$.

We refer to $\operatorname{Cov}(\widetilde{\epsilon}_l, \widetilde{\epsilon}_k)$ as the *temporal cross-covariance matrix*, since it measures cross-covariance between pseudogradient error vectors at optimization time steps k and l. When $k = l$, it reduces to the usual covariance matrix of $\widetilde{\epsilon}_k$.

Remarks. Intuitively, nontrivial temporal cross-covariance between $\widetilde{\epsilon}_k$ and $\widetilde{\epsilon}_l$ is possible, for instance, when

1. the parameter iterates change slowly so that the induced error process varies gradually and remains correlated across nearby steps, and
2. there is algorithmic dependence across steps, for example through overlapping or dependent mini-batch sampling.

Conversely, temporal cross-covariances may be small in settings where the pseudogradient errors decorrelate rapidly across steps, for example due to sufficiently randomized mini-batch sampling or other sources of algorithmic randomness that reduce the temporal dependence.

■

Corollary 5 (Total variance via the trace of the covariance of the iterate-estimation error) *Under the assumptions of Lemma 4,*

$$\operatorname{tr}(\operatorname{Cov}[\widetilde{e}]) = \gamma^2 \sum_{k=1}^{t} \sum_{l=1}^{t} a_{k,t} a_{l,t} \operatorname{tr}\left(\operatorname{Cov}(\widetilde{\epsilon}_l, \widetilde{\epsilon}_k)\right) \tag{54}$$

where $\operatorname{tr}(\operatorname{Cov}[\widetilde{e}])$ *is the total variance, i.e.,* $\operatorname{tr}(\operatorname{Cov}[\widetilde{e}]) = \sum_{i=1}^{d} \operatorname{Cov}[\widetilde{e}]_{i,i} = \sum_{i=1}^{d} \operatorname{Var}\left((\widetilde{e})_i\right)$. *Moreover, since* $\operatorname{Cov}[\widetilde{e}]$ *is a covariance matrix, it is positive semidefinite, and hence* $\operatorname{tr}(\operatorname{Cov}[\widetilde{e}]) \geq 0$.

Proof By Lemma 4 and linearity of trace,

$$\operatorname{tr}(\operatorname{Cov}[\widetilde{e}]) = \operatorname{tr}\left(\gamma^2 \sum_{k=1}^{t} \sum_{l=1}^{t} a_{k,t} a_{l,t} \operatorname{Cov}(\widetilde{\epsilon}_l, \widetilde{\epsilon}_k)\right) = \gamma^2 \sum_{k=1}^{t} \sum_{l=1}^{t} a_{k,t} a_{l,t} \operatorname{tr}\left(\operatorname{Cov}(\widetilde{\epsilon}_l, \widetilde{\epsilon}_k)\right)$$

Since $\operatorname{Cov}[\widetilde{e}]$ is a covariance matrix, it is positive semidefinite, hence $\operatorname{tr}(\operatorname{Cov}[\widetilde{e}]) \geq 0$. ■

Remark. Although $\operatorname{Cov}[\widetilde{e}]$ is positive semidefinite and thus $\operatorname{tr}(\operatorname{Cov}[\widetilde{e}]) \geq 0$, the cross-covariance matrices $\operatorname{Cov}(\widetilde{\epsilon}_l, \widetilde{\epsilon}_k)$ for $k \neq l$ need not be positive semidefinite, and $\operatorname{tr}(\operatorname{Cov}(\widetilde{\epsilon}_l, \widetilde{\epsilon}_k))$ may be any sign. We can write the trace identity that separates diagonal and off-diagonal indices to see the self-terms and cross-terms

$$\operatorname{tr}(\operatorname{Cov}[\widetilde{e}]) = \underbrace{\gamma^2 \sum_{k=1}^{t} a_{k,t}^2 \operatorname{tr}\left(\operatorname{Cov}(\widetilde{\epsilon}_k, \widetilde{\epsilon}_k)\right)}_{\text{self terms, where } \operatorname{tr}(\operatorname{Cov}(\widetilde{\epsilon}_k, \widetilde{\epsilon}_k)) = \sum_{i=1}^{d} \operatorname{Var}\left((\widetilde{\epsilon}_k)_i\right) \geq 0} \underbrace{+ \gamma^2 \sum_{\substack{k,l=1 \\ k \neq l}}^{t} a_{k,t} a_{l,t} \operatorname{tr}\left(\operatorname{Cov}(\widetilde{\epsilon}_l, \widetilde{\epsilon}_k)\right)}_{\text{cross terms, sign-indefinite}} \tag{55}$$

The first summation consists of self-covariance terms and is nonnegative, while the second summation captures temporal cross-covariance contributions and can have either sign.

Corollary 6 (Upper bound on total variance of the iterate-estimation error) *Under the assumptions of Corollary 5,*

$$\operatorname{tr}(\operatorname{Cov}[\widetilde{e}]) \leq \gamma^2 d \sum_{k=1}^{t} \sum_{l=1}^{t} \left| a_{k,t} a_{l,t} \right| \left\| \operatorname{Cov}(\widetilde{\epsilon}_l, \widetilde{\epsilon}_k) \right\|_{\mathrm{op}} \tag{56}$$

Proof By Corollary 5, we have

$$\operatorname{tr}(\operatorname{Cov}[\widetilde{e}]) = \gamma^2 \sum_{k=1}^{t} \sum_{l=1}^{t} a_{k,t} a_{l,t} \operatorname{tr}\big(\operatorname{Cov}(\widetilde{\epsilon}_l, \widetilde{\epsilon}_k)\big) \tag{57}$$

Although $\operatorname{tr}(\operatorname{Cov}[\widetilde{e}]) \geq 0$, the individual cross terms $\operatorname{tr}(\operatorname{Cov}(\widetilde{\epsilon}_l, \widetilde{\epsilon}_k))$ with $l \neq k$ can have either sign, so the summand in the double sum in (57) is not guaranteed to be nonnegative. Therefore, the standard first step is to control the magnitude of the sum by taking absolute values and applying the triangle inequality.

$$\begin{aligned}
\operatorname{tr}(\operatorname{Cov}[\widetilde{e}]) &\leq \left| \gamma^2 \sum_{k=1}^{t} \sum_{l=1}^{t} a_{k,t} a_{l,t} \operatorname{tr}\big(\operatorname{Cov}(\widetilde{\epsilon}_l, \widetilde{\epsilon}_k)\big) \right| && (58) \\
&= \gamma^2 \left| \sum_{k=1}^{t} \sum_{l=1}^{t} a_{k,t} a_{l,t} \operatorname{tr}\big(\operatorname{Cov}(\widetilde{\epsilon}_l, \widetilde{\epsilon}_k)\big) \right| \quad (\text{by homogeneity and since } \gamma^2 \geq 0) && (59) \\
&\leq \gamma^2 \sum_{k=1}^{t} \sum_{l=1}^{t} \left| a_{k,t} a_{l,t} \right| \left| \operatorname{tr}\big(\operatorname{Cov}(\widetilde{\epsilon}_l, \widetilde{\epsilon}_k)\big) \right| \quad (\text{by the triangle inequality}) && (60)
\end{aligned}$$

We now upper bound $\left| \operatorname{tr}(\operatorname{Cov}(\widetilde{\epsilon}_l, \widetilde{\epsilon}_k)) \right|$ by a matrix norm. This connection can be made because $\operatorname{tr}(A)$ can also be written as a Frobenius inner product. Concretely, for any square A, $\operatorname{tr}(A) = \operatorname{tr}(A^\top) = \operatorname{tr}(A^\top I_d) = \langle A, I_d \rangle_F$, where $\langle A, B \rangle_F := \operatorname{tr}(A^\top B)$.

Hence, let $C_{l,k} := \operatorname{Cov}(\widetilde{\epsilon}_l, \widetilde{\epsilon}_k) \in \mathbb{R}^{d \times d}$. Then

$$\begin{aligned}
\left| \operatorname{tr}(C_{l,k}) \right| &= \left| \operatorname{tr}(C_{l,k}^\top) \right| \quad (\text{since } \operatorname{tr}(A) = \operatorname{tr}(A^\top) \text{ for a square } A) \\
&= \left| \operatorname{tr}(C_{l,k}^\top I_d) \right| \\
&= \left| \langle C_{l,k}, I_d \rangle_F \right| \quad (\text{since } \langle A, B \rangle_F := \operatorname{tr}(A^\top B) \text{ for real valued matrices})
\end{aligned}$$

Then using the duality form of the nuclear norm, we can rescale B and take the supremum

$$\|A\|_* = \sup_{\|B\|_{\mathrm{op}} \leq 1} \left| \langle A, B \rangle_F \right| = \sup_{B \neq 0} \frac{\left| \langle A, B \rangle_F \right|}{\|B\|_{\mathrm{op}}} \qquad \text{for all } A \in \mathbb{R}^{d \times d}$$

and since $\|I_d\|_{\mathrm{op}} = 1$, we obtain

$$\left| \operatorname{tr}(C_{l,k}) \right| = \left| \langle C_{l,k}, I_d \rangle_F \right| \leq \sup_{\|B\|_{\mathrm{op}} \leq 1} \left| \langle C_{l,k}, B \rangle_F \right| = \|C_{l,k}\|_*$$

Substituting this into (60) yields

$$\operatorname{tr}(\operatorname{Cov}[\widetilde{e}]) \leq \gamma^2 \sum_{k=1}^{t} \sum_{l=1}^{t} \left| a_{k,t} a_{l,t} \right| \|C_{l,k}\|_* \tag{61}$$

Next, we can see the relation of the nuclear norm to the Frobenius norm by expanding in singular values. Fix any l, k and let $C_{l,k} \in \mathbb{R}^{d \times d}$ be a square matrix with singular values $\sigma_1(C_{l,k}), \ldots, \sigma_d(C_{l,k}) \geq$

0. We know that the nuclear norm $\|C_{l,k}\|_*$ is the sum of the singular values, whereas the Frobenius norm $\|C_{l,k}\|_F$ is the square root of the sum of squared singular values. This makes it natural to relate the sum to the root-sum-of-squares via Cauchy–Schwarz.

$$\begin{aligned}
\|C_{l,k}\|_* &= \sum_{i=1}^{d} \sigma_i(C_{l,k}) \quad (\text{since } C_{l,k} \in \mathbb{R}^{d\times d}, \text{ it has } d \text{ singular values}) \\
&= \langle (\sigma_1(C_{l,k}), \dots, \sigma_d(C_{l,k})),\, \mathbf{1}_d \rangle \\
&\le \|(\sigma_1(C_{l,k}), \dots, \sigma_d(C_{l,k}))\|_2 \, \|\mathbf{1}_d\|_2 \quad (\text{by Cauchy–Schwarz}) \\
&= \sqrt{\sum_{i=1}^{d} \sigma_i(C_{l,k})^2} \cdot \sqrt{d} \\
&= \sqrt{d}\, \|C_{l,k}\|_F \quad (\text{since } \|C_{l,k}\|_F^2 = \sum_{i=1}^{d} \sigma_i(C_{l,k})^2\,)
\end{aligned}$$

Substituting this into (61) yields

$$\operatorname{tr}(\operatorname{Cov}[\widetilde{e}]) \le \gamma^2 \sum_{k=1}^{t} \sum_{l=1}^{t} \left| a_{k,t} a_{l,t} \right| \sqrt{d}\, \|C_{l,k}\|_F \quad (\text{since } \|A\|_* \le \sqrt{d}\, \|A\|_F \text{ for } A \in \mathbb{R}^{d\times d}\,) \tag{62}$$

Finally, we connect the Frobenius norm to the operator norm. Both are singular-value norms: $\|C\|_F$ equals the 2-norm of the vector of singular values, while $\|C\|_{\mathrm{op}}$ is the largest singular value. Thus, we can bound the Frobenius norm in terms of the worst-case singular value:

$$\|C\|_F^2 = \sum_{i=1}^{d} \sigma_i(C)^2 \le \sum_{i=1}^{d} \big(\max_j \sigma_j(C) \big)^2 = d\, \|C\|_{\mathrm{op}}^2$$

hence $\|C\|_F \le \sqrt{d}\, \|C\|_{\mathrm{op}}$ for any $C \in \mathbb{R}^{d\times d}$. Substituting this into (62) yields

$$\begin{aligned}
\operatorname{tr}(\operatorname{Cov}[\widetilde{e}]) &\le \gamma^2 \sum_{k=1}^{t} \sum_{l=1}^{t} \left| a_{k,t} a_{l,t} \right| \sqrt{d} \left(\sqrt{d}\, \|C_{l,k}\|_{\mathrm{op}} \right) && (63) \\
&= \gamma^2 d \sum_{k=1}^{t} \sum_{l=1}^{t} \left| a_{k,t} a_{l,t} \right| \left\| \operatorname{Cov}(\widetilde{\epsilon}_l, \widetilde{\epsilon}_k) \right\|_{\mathrm{op}} && (64)
\end{aligned}$$

■

Corollary 7 (Total variance of the iterate-estimation error under vanishing temporal cross-covariance) *Assume the setting of Corollary 5. If, in addition,*

$$\operatorname{Cov}(\widetilde{\epsilon}_l, \widetilde{\epsilon}_k) = 0_{d\times d} \qquad \textit{for all } k \ne l$$

define

$$\mathrm{V} := \gamma^2 \sum_{k=1}^{t} a_{k,t}^2 \operatorname{tr}\left(\operatorname{Cov}(\widetilde{\epsilon}_k, \widetilde{\epsilon}_k) \right) \tag{65}$$

Then

$$\mathrm{tr}(\mathrm{Cov}[\widetilde{e}]) = \mathrm{V} \tag{66}$$

Proof By Corollary 5, we have the decomposition

$$\mathrm{tr}(\mathrm{Cov}[\widetilde{e}]) = \gamma^2 \sum_{k=1}^{t} a_{k,t}^2 \,\mathrm{tr}\left(\mathrm{Cov}(\widetilde{\epsilon}_k, \widetilde{\epsilon}_k)\right) + \gamma^2 \sum_{\substack{k,l=1 \\ k \neq l}}^{t} a_{k,t} a_{l,t} \,\mathrm{tr}\left(\mathrm{Cov}(\widetilde{\epsilon}_l, \widetilde{\epsilon}_k)\right)$$

Under the stated assumption, $\mathrm{Cov}(\widetilde{\epsilon}_l, \widetilde{\epsilon}_k) = 0_{d\times d}$ for all $k \neq l$, so the entire cross-term summation vanishes. Therefore,

$$\mathrm{tr}(\mathrm{Cov}[\widetilde{e}]) = \gamma^2 \sum_{k=1}^{t} a_{k,t}^2 \,\mathrm{tr}\left(\mathrm{Cov}(\widetilde{\epsilon}_k, \widetilde{\epsilon}_k)\right) = \mathrm{V}$$

■

Corollary 8 (Upper bound on total variance under vanishing temporal cross-covariance) *Assume the setting of Corollary 7, where*

$$\mathrm{Cov}(\widetilde{\epsilon}_l, \widetilde{\epsilon}_k) = 0_{d\times d} \qquad \text{for all } k \neq l$$

then

$$V := \mathrm{tr}(\mathrm{Cov}[\widetilde{e}]) = \gamma^2 \sum_{k=1}^{t} a_{k,t}^2 \,\mathrm{tr}\left(\mathrm{Cov}(\widetilde{\epsilon}_k, \widetilde{\epsilon}_k)\right) \tag{67}$$

and, in particular,

$$V \leq \gamma^2 d \sum_{k=1}^{t} a_{k,t}^2 \left\| \mathrm{Cov}(\widetilde{\epsilon}_k, \widetilde{\epsilon}_k) \right\|_{\mathrm{op}} \tag{68}$$

Proof This proof begins from the same trace expansion as Corollary 5. Under the assumption of Corollary 7 that $\mathrm{Cov}(\widetilde{\epsilon}_k, \widetilde{\epsilon}_l) = 0_{d\times d}$ for all $k \neq l$, all cross terms vanish, leaving only diagonal self-covariances, and we then apply a bound on the trace of the PSD covariance matrix $\mathrm{Cov}(\widetilde{\epsilon}_k, \widetilde{\epsilon}_k) \succeq 0$.

By Corollary 7, under the stated assumption all terms with $k \neq l$ vanish, hence

$$V := \mathrm{tr}(\mathrm{Cov}[\widetilde{e}]) = \gamma^2 \sum_{k=1}^{t} a_{k,t}^2 \,\mathrm{tr}\left(\mathrm{Cov}(\widetilde{\epsilon}_k, \widetilde{\epsilon}_k)\right)$$

Moreover, for each k, the covariance matrix $\mathrm{Cov}(\widetilde{\epsilon}_k, \widetilde{\epsilon}_k) \in \mathbb{R}^{d\times d}$ is positive semidefinite. Therefore, by the spectral theorem, there exist an orthogonal matrix $Q \in \mathbb{R}^{d\times d}$ and a diagonal matrix $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_d)$ with $\lambda_i \geq 0$ such that

$$\mathrm{Cov}(\widetilde{\epsilon}_k, \widetilde{\epsilon}_k) = Q \Lambda Q^\top$$

Thus

$$\begin{aligned}
\operatorname{tr}\big(\operatorname{Cov}(\widetilde{\epsilon}_k, \widetilde{\epsilon}_k)\big) &= \operatorname{tr}(Q\Lambda Q^\top) \quad (\text{by the spectral decomposition}) \\
&= \operatorname{tr}(\Lambda Q^\top Q) \quad (\text{by cyclic property of trace}) \\
&= \operatorname{tr}(\Lambda) \quad (\text{since } Q^\top Q = I_d) \\
&= \sum_{i=1}^{d} \lambda_i\big(\operatorname{Cov}(\widetilde{\epsilon}_k, \widetilde{\epsilon}_k)\big) \quad (\text{since } \Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_d) \text{ contains the eigenvalues of } \operatorname{Cov}(\widetilde{\epsilon}_k, \widetilde{\epsilon}_k)) \\
&\le \sum_{i=1}^{d} \lambda_{\max}\big(\operatorname{Cov}(\widetilde{\epsilon}_k, \widetilde{\epsilon}_k)\big) \\
&= d\,\lambda_{\max}\big(\operatorname{Cov}(\widetilde{\epsilon}_k, \widetilde{\epsilon}_k)\big) \\
&= d\,\big\|\operatorname{Cov}(\widetilde{\epsilon}_k, \widetilde{\epsilon}_k)\big\|_{\mathrm{op}}
\end{aligned}$$

Multiplying by $\gamma^2 a_{k,t}^2$ and summing over k yields

$$V \le \gamma^2 d \sum_{k=1}^{t} a_{k,t}^2 \big\|\operatorname{Cov}(\widetilde{\epsilon}_k, \widetilde{\epsilon}_k)\big\|_{\mathrm{op}}$$

■

Theorem 9 (Projected variance of the iterate-estimation error and a general upper bound) *Fix any unit vector $u \in \mathbb{R}^d$ with $\|u\|_2 = 1$. Let $\widetilde{e} \in \mathbb{R}^d$ denote the iterate-estimation error as defined previously, and recall from Lemma 2 that*

$$\widetilde{e} = -\gamma \sum_{k=1}^{t} a_{k,t} \widetilde{\epsilon}_k$$

where $\widetilde{\epsilon}_k \in \mathbb{R}^d$ denotes the pseudogradient error at step k. Define $a_t := [a_{1,t}, \ldots, a_{t,t}]^\top \in \mathbb{R}^t$ and define the stacked cross-covariance matrix $\hat{\Sigma} \in \mathbb{R}^{td \times td}$ by its (l, k) block entries

$$\hat{\Sigma}_{l,k} := \operatorname{Cov}(\widetilde{\epsilon}_l, \widetilde{\epsilon}_k) \in \mathbb{R}^{d \times d}$$

Then the projected variance admits the quadratic-form representation

$$\operatorname{Var}[u^\top \widetilde{e}] = \gamma^2 (a_t \otimes u)^\top \hat{\Sigma} (a_t \otimes u) \tag{69}$$

Moreover, the following general upper bound becomes

$$\operatorname{Var}[u^\top \widetilde{e}] \le \gamma^2 \|\hat{\Sigma}\|_{\mathrm{op}} \|a_t\|_2^2 \tag{70}$$

Proof A standard way to quantify the iterate-estimation error is to study its variance along arbitrary directions. Since $\widetilde{e} \in \mathbb{R}^d$ is a vector-valued random variable, we consider its one-dimensional projection onto an arbitrary unit vector $u \in \mathbb{R}^d$ with $\|u\|_2 = 1$. The orthogonal projection of $\widetilde{e}$ onto $\operatorname{span}(u)$ is $P_{\operatorname{span}(u)}(\widetilde{e}) = \langle \widetilde{e}, u \rangle u$, where the scalar projection coefficient is $\langle \widetilde{e}, u \rangle$. We quantify the deviation along direction u via the projected variance, which can be expressed in the standard quadratic-form form.

$$\mathrm{Var}[u^\top \widetilde{e}] = u^\top \, \mathrm{Cov}[\widetilde{e}]u \tag{71}$$

Substituting Lemma 4 yields

$$\mathrm{Var}[u^\top \widetilde{e}] = u^\top \Big(\gamma^2 \sum_{k=1}^{t}\sum_{l=1}^{t} a_{k,t}a_{l,t}\,\mathrm{Cov}(\widetilde{\epsilon}_l, \widetilde{\epsilon}_k)\Big)u \tag{72}$$

$$= \gamma^2 \sum_{k=1}^{t}\sum_{l=1}^{t} a_{k,t}a_{l,t}\, u^\top\,\mathrm{Cov}(\widetilde{\epsilon}_l, \widetilde{\epsilon}_k)u \quad \text{(by distributivity and homogeneity)} \tag{73}$$

$$= \gamma^2 \sum_{k=1}^{t}\sum_{l=1}^{t} a_{k,t}a_{l,t}\Big(\sum_{i=1}^{d}\sum_{j=1}^{d} u_i u_j\ \mathrm{Cov}(\widetilde{\epsilon}_l, \widetilde{\epsilon}_k)_{j,i}\Big) \tag{74}$$

$$\text{(by the coordinate expansion of the quadratic form)} \tag{75}$$

$$= \gamma^2 \sum_{k=1}^{t}\sum_{l=1}^{t}\sum_{i=1}^{d}\sum_{j=1}^{d}(a_{k,t}u_i)(a_{l,t}u_j)\ \mathrm{Cov}(\widetilde{\epsilon}_l, \widetilde{\epsilon}_k)_{j,i} \tag{76}$$

We observe that the final expression, involving the coefficient products $(a_{l,t}u_j)$ and $(a_{k,t}u_i)$ and the matrix entries $\mathrm{Cov}(\widetilde{\epsilon}_l, \widetilde{\epsilon}_k)_{j,i}$, matches the coordinate expansion of a quadratic form. This suggests that we can package it as a quadratic form by reindexing the pair (l, j) into a single index and stacking the corresponding coefficients.

Define the reindexed quantities

$$x_{(l,j)} := a_{l,t}u_j, \qquad x_{(k,i)} := a_{k,t}u_i, \qquad \hat{\Sigma}_{(l,j),(k,i)} := \mathrm{Cov}(\widetilde{\epsilon}_l, \widetilde{\epsilon}_k)_{j,i}$$

Then the previous expression takes the form of a quadratic form under the index set $\mathcal{I} := \{1, \ldots, t\} \times \{1, \ldots, d\}$,

$$\mathrm{Var}[u^\top \widetilde{e}] = \gamma^2 \sum_{(l,j)\in\mathcal{I}}\sum_{(k,i)\in\mathcal{I}} x_{(l,j)}\hat{\Sigma}_{(l,j),(k,i)}x_{(k,i)} \tag{77}$$

Equivalently, in stacked-vector notation, let

$$u := \begin{bmatrix} u_1 \\ \vdots \\ u_d \end{bmatrix} \in \mathbb{R}^d, \qquad a_t := \begin{bmatrix} a_{1,t} \\ \vdots \\ a_{t,t} \end{bmatrix} \in \mathbb{R}^t, \qquad x := \begin{bmatrix} a_{1,t}u \\ \vdots \\ a_{t,t}u \end{bmatrix} \in \mathbb{R}^{td}$$

and define the block matrix $\hat{\Sigma} \in \mathbb{R}^{td\times td}$ by

$$\hat{\Sigma} := \begin{bmatrix} \mathrm{Cov}(\widetilde{\epsilon}_1, \widetilde{\epsilon}_1) & \cdots & \mathrm{Cov}(\widetilde{\epsilon}_1, \widetilde{\epsilon}_t) \\ \vdots & \ddots & \vdots \\ \mathrm{Cov}(\widetilde{\epsilon}_t, \widetilde{\epsilon}_1) & \cdots & \mathrm{Cov}(\widetilde{\epsilon}_t, \widetilde{\epsilon}_t) \end{bmatrix}$$

With these definitions, we can rewrite the projected variance as

$$\mathrm{Var}[u^\top \widetilde{e}] = \gamma^2 x^\top \hat{\Sigma} x \tag{78}$$

Moreover, defining $a_t := [a_{1,t}, \ldots, a_{t,t}]^\top$, we can conveniently rewrite the stacked vector x as the Kronecker product $x = a_t \otimes u$ to reveal a_t and u more explicitly.

$$\mathrm{Var}[u^\top \widetilde{e}] = \gamma^2 (a_t \otimes u)^\top \hat{\Sigma} (a_t \otimes u) \tag{79}$$

We now bound the projected variance,

$$\begin{aligned}
\mathrm{Var}[u^\top \widetilde{e}] &= \gamma^2 (a_t \otimes u)^\top \hat{\Sigma} (a_t \otimes u) && (80)\\
&\le \left| \gamma^2 (a_t \otimes u)^\top \hat{\Sigma} (a_t \otimes u) \right| && (81)\\
&= |\gamma^2| \ \left| (a_t \otimes u)^\top \hat{\Sigma} (a_t \otimes u) \right| \quad (\text{ by homogeneity }) && (82)\\
&= \gamma^2 \left| (a_t \otimes u)^\top \hat{\Sigma} (a_t \otimes u) \right| \quad (\text{ since } \gamma^2 \ge 0\,) && (83)\\
&= \gamma^2 \left| \langle a_t \otimes u,\ \hat{\Sigma}(a_t \otimes u) \rangle \right| \quad (\text{ Euclidean inner product }) && (84)\\
&\le \gamma^2 \|a_t \otimes u\|_2 \ \|\hat{\Sigma}(a_t \otimes u)\|_2 \quad (\text{ by Cauchy–Schwarz }) && (85)
\end{aligned}$$

Intuitively, since the stacked cross-covariance matrix $\hat{\Sigma}$ depends on the training randomness, its structure can vary. We therefore want to control how much $\hat{\Sigma}$ can amplify an arbitrary direction. This worst-case effect can be quantified by the operator norm.

By the definition of the operator norm, for any nonzero vector $v \in \mathbb{R}^{td}$,

$$\|\hat{\Sigma}\|_{\mathrm{op}} := \sup_{v \neq 0_{td}} \frac{\|\hat{\Sigma} v\|_2}{\|v\|_2}$$

Applying this with $v = a_t \otimes u \neq 0_{td}$ yields

$$\begin{aligned}
\mathrm{Var}[u^\top \widetilde{e}] &\le \gamma^2 \|a_t \otimes u\|_2 \left(\frac{\|\hat{\Sigma}(a_t \otimes u)\|_2}{\|a_t \otimes u\|_2} \right) \|a_t \otimes u\|_2 && (86)\\
&\le \gamma^2 \|a_t \otimes u\|_2 \ \|\hat{\Sigma}\|_{\mathrm{op}} \ \|a_t \otimes u\|_2 && (87)\\
&= \gamma^2 \|\hat{\Sigma}\|_{\mathrm{op}} \ \|a_t \otimes u\|_2^2 && (88)
\end{aligned}$$

To reveal a_t and u more explicitly for analysis,

$$\mathrm{Var}[u^\top \tilde{e}] \leq \gamma^2 \|\hat{\Sigma}\|_{\mathrm{op}} \; \|a_t \otimes u\|_2^2 \tag{89}$$

$$= \gamma^2 \|\hat{\Sigma}\|_{\mathrm{op}} \left\| \begin{matrix} a_{1,t}u \\ a_{2,t}u \\ \vdots \\ a_{t,t}u \end{matrix} \right\|_2^2 \tag{90}$$

$$= \gamma^2 \|\hat{\Sigma}\|_{\mathrm{op}} \sum_{i=1}^{t} \sum_{j=1}^{d} (a_{i,t} u_j)^2 \tag{91}$$

$$= \gamma^2 \|\hat{\Sigma}\|_{\mathrm{op}} \sum_{i=1}^{t} \sum_{j=1}^{d} a_{i,t}^2 u_j^2 \tag{92}$$

$$= \gamma^2 \|\hat{\Sigma}\|_{\mathrm{op}} \sum_{i=1}^{t} a_{i,t}^2 \Big(\sum_{j=1}^{d} u_j^2 \Big) \quad (\text{factorize the constant}) \tag{93}$$

$$= \gamma^2 \|\hat{\Sigma}\|_{\mathrm{op}} \sum_{i=1}^{t} a_{i,t}^2 \|u\|_2^2 \quad (\text{by the definition of the squared 2-norm}) \tag{94}$$

$$= \gamma^2 \|\hat{\Sigma}\|_{\mathrm{op}} \|u\|_2^2 \sum_{i=1}^{t} a_{i,t}^2 \quad (\text{factor out the constant}) \tag{95}$$

$$= \gamma^2 \|\hat{\Sigma}\|_{\mathrm{op}} \underbrace{\|u\|_2^2}_{=1} \|a_t\|_2^2 \quad (\text{by the definition of the squared 2-norm}) \tag{96}$$

$$= \gamma^2 \|\hat{\Sigma}\|_{\mathrm{op}} \|a_t\|_2^2 \tag{97}$$

Interpretation:

1. $\|\hat{\Sigma}\|_{\mathrm{op}}$ quantifies the operator-norm magnitude of the stacked cross-covariance matrix of the pseudogradient errors across time steps. It captures the worst-case strength of the temporal dependence and variability in the error process, induced by the training dynamics, and is driven by stochasticity rather than being directly tunable in closed form.

2. $\|a_t\|_2^2$ aggregates the squared Nesterov weighting coefficients in the unrolled outer update, and therefore quantifies the extent to which long-memory weighting can amplify the contribution of pseudogradient errors to the projected variance.

■

2.1. *Error amplifiers*

We can understand this effect by examining the scalar weighting coefficients $a_{k,t} \in \mathbb{R}$ and $a_{l,t} \in \mathbb{R}$ that appear in the unrolled Nesterov outer update.

It is not surprising that these coefficients exhibit a monotone dependence on the time gap $t - k$, since the Nesterov outer update is generated by a linear momentum recursion with parameter $\mu \in$

$[0, 1)$. We expect unrolling such recurrence produces geometrically weighted contributions from past steps. To make this dependence explicit in our setting, we analyze the closed-form coefficient $a_{k,t}$ directly.

In particular, we show that the time gap $t - k$ affects $a_{k,t}$ through the second summand in the closed-form unrolling. For fixed t, a smaller index k corresponds to an older momentum-memory term, which allows us to study the monotonicity of $a_{k,t}$ as a function of the gap $t - k$.

Theorem 10 (Error-amplifying memory) *Fix $t \geq 1$ and parameters $\mu \in [0, 1)$ and $\tau \in [0, 1]$. Let $a_{k,t} \in \mathbb{R}$ denote the Nesterov weighting coefficient in the unrolled outer update, defined by*

$$a_{k,t} := 1 + \frac{(1-\tau)\big(\mu - \mu^{t-k+2}\big)}{1-\mu}, \qquad k \in \{1, \dots, t\}$$

Then $a_{k,t}$ is nonincreasing in k and nondecreasing in the time gap $d := t - k$. This means, older memory terms (smaller k, larger $t - k$) receive larger weights.

Proof Recall the Nesterov coefficient

$$a_{k,t} := 1 + \frac{(1-\tau)\big(\mu - \mu^{t-k+2}\big)}{1-\mu}$$

Let $d := t - k$ denote the time-gap, where $d \in \{0, 1, \cdots, t-1\}$.

Here we focus on the second summand since it is the only term that varies with k. Substituting $d = t - k$ will give us

$$a_{k,t} = 1 + \frac{(1-\tau)\big(\mu - \mu^{d+2}\big)}{1-\mu}$$

Since $d + 2$ contains a common fixed offset across successive gaps, we just relabel and define $n := d + 2 = t - k + 2$. Then

$$a_{k,t} = 1 + \frac{(1-\tau)\big(\mu - \mu^{n}\big)}{1-\mu}$$

Now we take the difference of successive increments in the gap. Let the second summand be the function of n

$$b_n := \frac{(1-\tau)}{1-\mu}\big(\mu - \mu^n\big)$$

so that $a_{k,t} = 1 + b_n$. Then

$$\begin{aligned}
b_{n+1} - b_n &= \frac{(1-\tau)}{1-\mu}\big(\mu - \mu^{n+1}\big) - \frac{(1-\tau)}{1-\mu}\big(\mu - \mu^n\big) && (98)\\
&= \frac{(1-\tau)}{1-\mu}\big(-\mu^{n+1} + \mu^n\big) && (99)\\
&= \frac{(1-\tau)}{1-\mu}\mu^n(1-\mu) && (100)\\
&= (1-\tau)\mu^n \quad (\text{cancel } (1-\mu)) && (101)
\end{aligned}$$

Since $\tau \in [0, 1]$ and $\mu \in [0, 1)$, we have $(1-\tau)\mu^n \geq 0$. Therefore $b_{n+1} \geq b_n$, so b_n is nondecreasing in n, and hence $a_{k,t} = 1 + b_n$ is also nondecreasing in n.

Finally, $n = t-k+2$ increases when k decreases. Equivalently, for fixed t, $a_{k,t}$ is nonincreasing in k and nondecreasing in the gap $t-k$. In particular, smaller k (older memory terms) receive larger coefficients.

This is reflected in the covariance expansion in Lemma 4

$$\gamma^2 \sum_{l=1}^{t} \sum_{k=1}^{t} \underbrace{a_{l,t} a_{k,t}}_{\text{error-amplifying coefficients}} \underbrace{\mathrm{Cov}(\widetilde{\epsilon}_l, \widetilde{\epsilon}_k)}_{\text{temporal cross-covariance of errors}}$$

which shows that long-memory weighting can magnify the contribution of earlier pseudogradient errors. This motivates mechanisms that control the effective memory horizon (e.g., truncation via periodic restarts). ■

2.2. Discussion on potential solutions for error amplification

1. Reweighting. If one could estimate the pseudogradient error at specific time steps, then one could in principle reweight the geometric contributions in the outer momentum to dampen the influence of steps with large error magnitude or high variance. This would require a method to estimate such error magnitudes in the absence of per-step synchronized gradients.

2. Periodic momentum restarts. A simple way is to truncate the effective memory while still leveraging the benefits of outer momentum. In practice, this can be implemented via periodic *restarts*, where the outer momentum state is periodically reset, which maintains a sliding window of relevant history. This is the approach we explore in detail in this work.

References